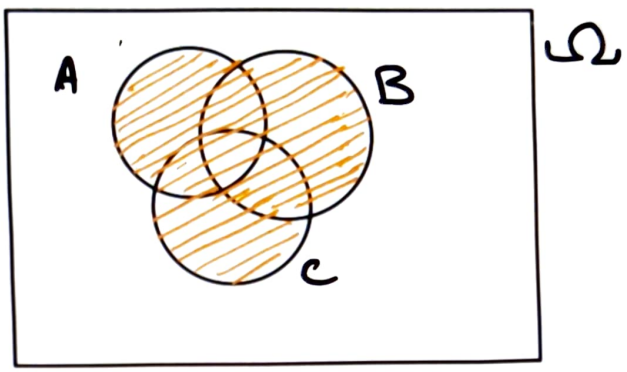
Problem set 1.

6.041 : Probabilistic Systems Analysis

1. (a) At least one of the events A, B, C occurs : $A \cup B \cup C$

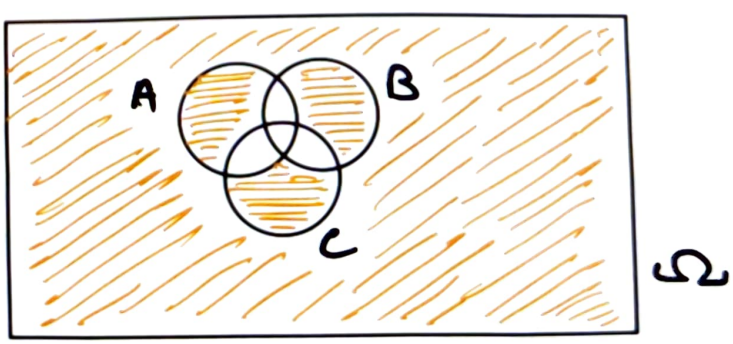


(b) At most one of the events A, B, C occurs :

$$(A \cup B \cup C)^c \cup (A^c \cup B \cup C)^c \cup (A \cup B^c \cup C)^c \cup (A \cup B \cup C^c)^c$$

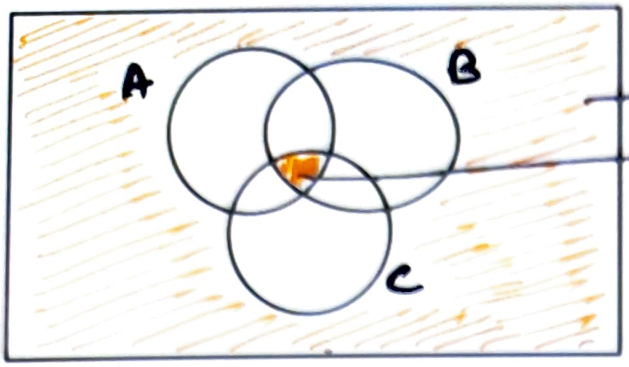
$(A \cup B \cup C)^c$: none of the events occurs

$(A^c \cup B \cup C)^c \cup (A \cup B^c \cup C)^c \cup (A \cup B \cup C^c)^c$: exactly one event occurs



(c) None of the events A, B, C occurs : $(A \cup B \cup C)^c$

(d) All the three events A, B, C occur : $A \cap B \cap C$

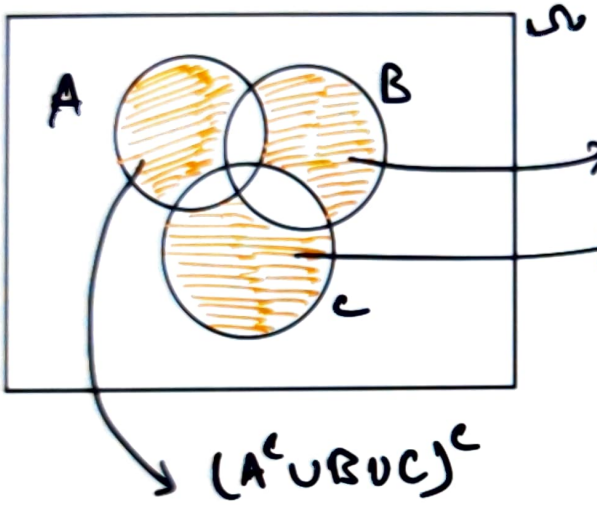


(c) $(A \cup B \cup C)^c$

(d) $A \cap B \cap C$

(e) Exactly one of the event A, B, C occurs:

$(A^c \cup B \cup C)^c \cup (A \cup B^c \cup C)^c \cup (A \cup B \cup C^c)^c$



$(A \cup B^c \cup C)^c$

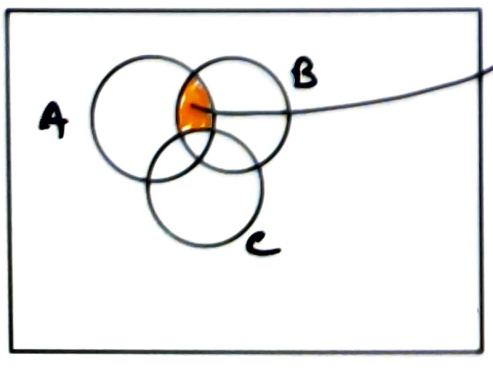
$(A \cup B \cup C^c)^c$

$(A^c \cup B \cup C)^c$

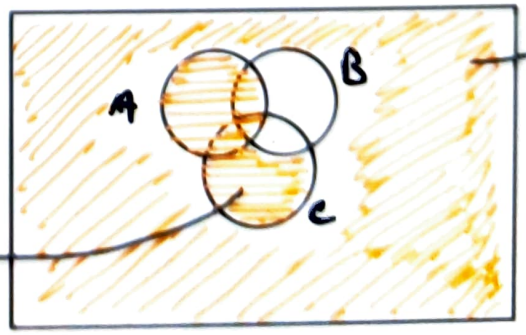
$A \cup (C \cap B^c) \cup (A \cup B \cup C)^c$

(f) Events A and B occur but not C : $(A \cap B) \cap C^c$

(g) Either event A occurs OR, if not, then B also does not occur : $A \cup (C \cap B^c) \cup (A \cup B \cup C)^c$



$(A \cap B) \cap C^c$



$A \cup (C \cap B^c)$

$(A \cup B \cup C)^c$

2. (a) $P(HHHT) = 1/8$

(b) $P(HTTH) = 1/8$

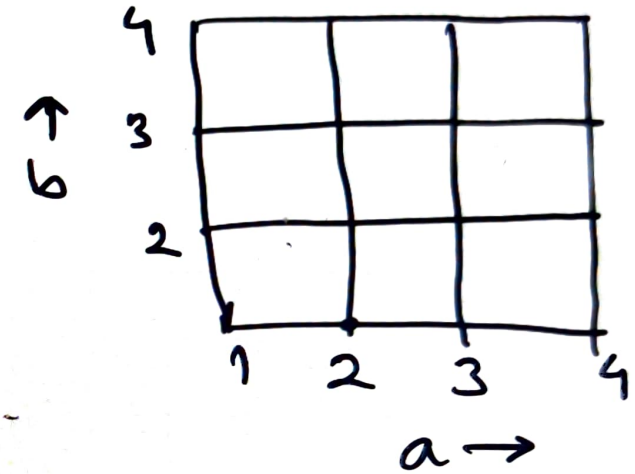
(c) $P(\{HHT, HTH, THH\}) = 3/8$

(d) $P(\{HHT, HTH, THH\}) = 3/8$

3. Let a = number on 1st die

b = number on 2nd die

$P((a,b)) = \alpha(a+b)$



$P(\Omega) = 1$

$= \alpha \sum_{a,b} a+b$

$= \alpha\,(2+2\times3+3\times4+4\times5+3\times6+2\times7+8)$

$= \alpha\,(2+6+12+20+18+14+8) = 80\alpha \Rightarrow \alpha = \frac{1}{80}$

(a) Even sum : $\{(1,1), (1,3), (3,1), (2,2), (2,4), (3,3), (4,2), (4,4)\}$

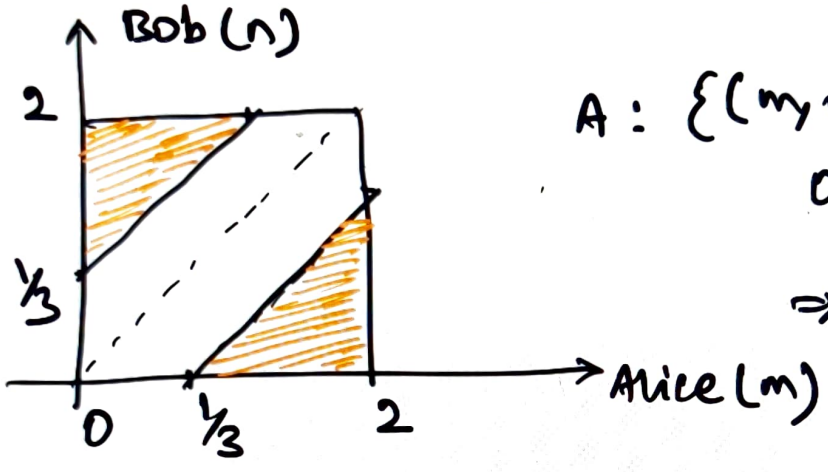
Probability of even sum $= \frac{2}{80} + \frac{3\times4}{80} + \frac{3\times6}{80} + \frac{8}{80}$

$= 1/2$

(b) $P(\{(2,3),(3,2)\}) = P(\{(2,3)\}) +$

$P(\{(3,2)\}) = \frac{2 \times 5}{80} = \frac{1}{8}$.

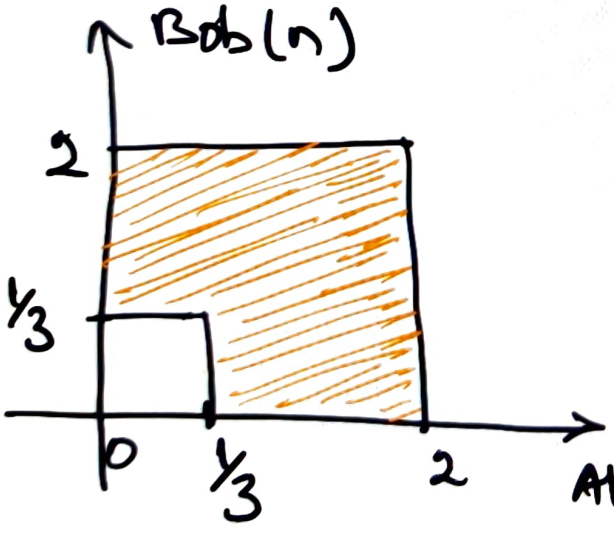
4.



$A: \{(m,n) \mid |m-n| > 1/3,$
$0 \leq m,n \leq 2\}$

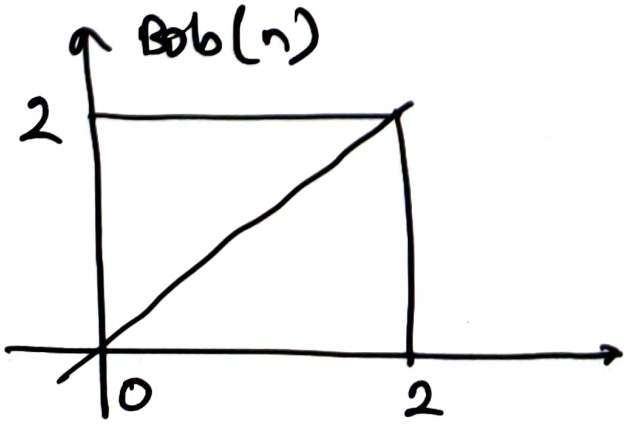
$\Rightarrow P(A) = 25/36$

Let m = Alice's number, n = Bob's number



$B: \{(m,n) \mid 1/3 \leq m,n \leq 2\}$

$\Rightarrow P(B) = \frac{35}{36}$



$C: \{(m,n) \mid 0 \leq m,n \leq 2,$
$m = n\}$

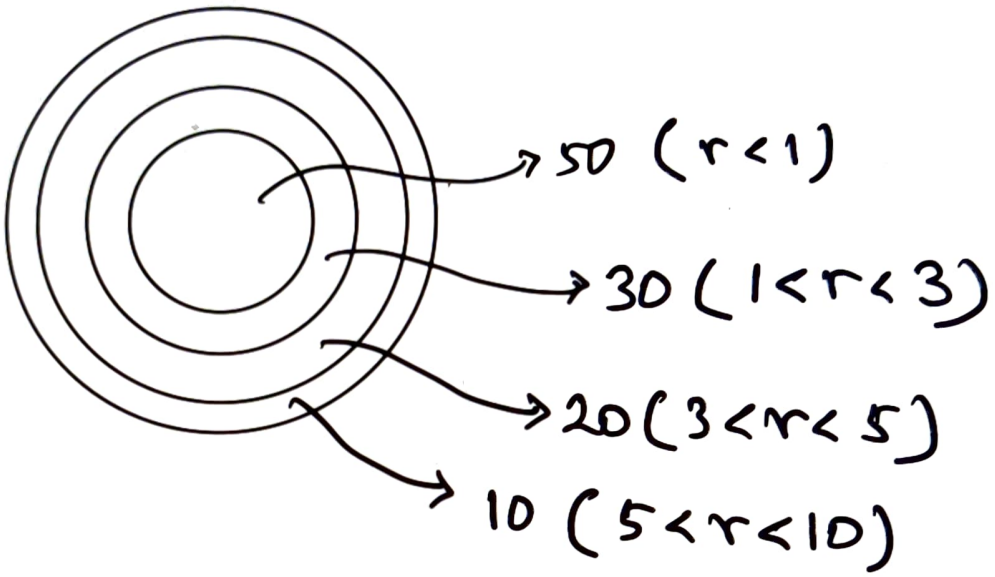
$\Rightarrow P(C) = 0$

→ no notion of area for a line

$D: \{(m,n) \mid m > \frac{1}{3},\ m \leq 2,\ 0 \leq n \leq 2\}$

$A \cap D = D \Rightarrow P(A \cap D) = \frac{1}{2} P(A) = \frac{25}{72}$

5.



(a) $P(\text{50 points in one throw}) = \frac{\pi(1)^2}{\pi(10)^2} = 0.01$

(b) $P(\text{30 points in 1 throw}) = \frac{\pi(3^2-1^2)}{\pi(10)^2} = 0.08$

(c) For John, $P(\text{right}) = 2 \times P(\text{left})$

and $P(\text{right}) + P(\text{left}) = 1$

$\Rightarrow P(\text{left}) = \frac{1}{3},\ P(\text{right} = \frac{2}{3})$

Let A = John scores 50 in 1 throw

B = John scores 30 in 1 throw

$$P(A) = P(A|\text{left})\,P(\text{left}) + P(A|\text{right})\,P(\text{right})$$

$$= 0.01 \times \tfrac{1}{3} + 0.01 \times \tfrac{2}{3} = 0.01$$

Note $P(A|\text{left}) = P(A|\text{right}) = 0.01$

Similarly $P(B|\text{left}) = P(B|\text{right}) = 0.00$

because all areas are halved.

$$P(B) = P(B|\text{left})\,P(\text{left}) + P(B|\text{right})\,P(\text{right})$$

$$= 0.08 \times \tfrac{1}{3} + 0.08 \times \tfrac{2}{3} = 0.08$$

G1. (a)



$$\lim_{n\to\infty} a_n = a \quad \text{and} \quad \lim_{n\to\infty} b_n = b$$

$$[a_{n+1}, b_{n+1}] \subset [a_n, b_n] \subset \ldots \subset [a_1, b_1]$$

$$\Rightarrow \bigcap_{i=1}^{n} [a_i, b_i] = [a_n, b_n]$$

$$\Rightarrow \lim_{n\to\infty} \bigcap_{i=1}^{n} [a_i, b_i] = \lim_{n\to\infty} [a_n, b_n] = [a, b]$$

$\Rightarrow \lim_{n\to\infty} P([a_n, b_n]) = P([a, b])$

(b)



$\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$

$[a_1, b_1] \subset [a_2, b_2] \subset \ldots \subset [a_n, b_n]$

$\Rightarrow \bigcup_{i=1}^{\infty} [a_i, b_i] = [a_n, b_n]$

$\Rightarrow \lim_{n\to\infty} \bigcup_{i=1}^{\infty} [a_i, b_i] = \lim_{n\to\infty} [a_n, b_n] = [a, b]$

$\Rightarrow \lim_{n\to\infty} P([a_n, b_n]) = P([a, b])$

6. $P(A \cap B \cap C) \geq P(A) + P(B) + P(C) - 2$

$1 - P(A \cap B \cap C) = P((A \cap B \cap C)^c)$

$= P(A^c \cup B^c \cup C^c)$

$\leq P(A^c) + P(B^c) + P(C^c)$

$= 1 - P(A) + 1 - P(B) + 1 - P(C)$

$\Rightarrow P(A) + P(B) + P(C) - 2 \leq P(A \cap B \cap C)$

Connections required :

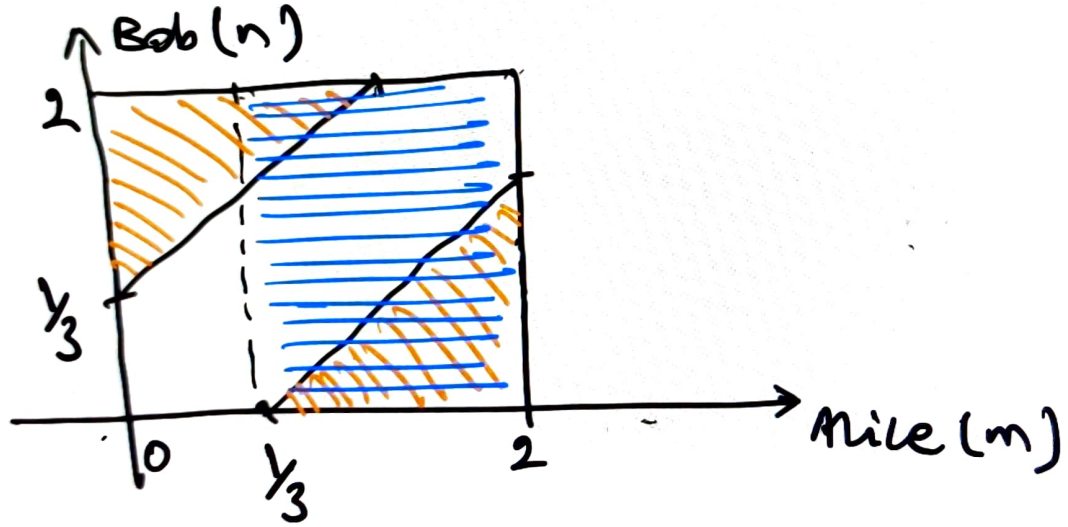
2(d) consider the case with no tails.

4. $P(A \cap D)$ → draw the area. It will take a part of the top triangle too.

2(d) More heads than tails :

$P(\{HHHH, HTH, THH, HHT\}) = 4/8 = 1/2$

4.



$A : \{(m,n) \mid |m-n| > 1/3,\ 0 \le m,n \le 2\}$

$D : \{(m,n) \mid 1/3 < m \le 2,\ 0 \le n \le 2\}$

Area of $A \cap D = 1/2 \times 5/3 \times 5/3 + 1/2 \times 4/3 \times 4/3$

$= 41/18$

$\Rightarrow P(A \cap D) = 41/72$